Chapter 1

Foundations of Clinical Sonography

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Terminology

- Terms that describe instrumentation: *Diagnostic medical ultrasound, ultrasound,* and *ultrasonography*
- Sonography: Visualizes soft-tissue structures of the body
- Echocardiography or echo: Sonographic examination of cardiac structures

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Terminology (Cont.)

- Sonographer: Allied health professional
  - Specialized education in ultrasound
  - Successful completion of national boards (American Registry of Diagnostic Medical Sonography)
- Sonologist: Physician (medical doctor)
  - Specialized training in ultrasound
  - Successful completion of national boards in his or her respective specialty

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Clinical Applications of Ultrasound

- General (abdominal, renal, and retroperitoneal, chest)
- Superficial (breast, thyroid, scrotum)
- Neonatal and Pediatric (abdomen, renal, hips, brain, spine)
- Cardiovascular (heart, vascular system)
- Interventional and Therapeutic-guided
- Obstetric and Gynecologic
- Intraoperative
- Musculoskeletal
- Point of Care

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Advantages of Ultrasound

- No harmful effects
- No ionizing radiation
- Portable equipment
- Real-time format

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Advantages of Ultrasound (Cont.)

- Multiplanar imaging: can “follow” the path of a tortuous vessel, moving cardiac structure, or fetus
- Doppler techniques: Allow qualitative and quantitative evaluation of blood flow
- Image enhancement contrast applications
- Inexpensive compared to other imaging systems

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Contrast Agents

- Ingested or administered intravenously
- Facilitates detection/diagnosis of specific pathologic conditions
- Early attempts to produce contrast effect involved administration of aerated saline or carbon dioxide
- Today’s research focuses on gas microspheres injected into the patient via IV access

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Role of the Sonographer

- Works directly with physicians and patients
- Interacts with nurses and other members of the medical staff
- Reviews patient's records to assess clinical history and symptoms; interprets laboratory values; should understand other diagnostic examinations
- Understands and operates complex ultrasound instrumentation

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How Is the Sonographer Different from Other Allied Health Professionals?

- Talks directly with patients to identify which symptoms relate directly to the examination
- Explains procedure to patient, and performs examination using the protocol established by the department
- Analyzes each image and correlates it to patient clinical history

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How Is the Sonographer Different from Other Allied Health Professionals? (Cont.)

- Uses independent judgment about needing to make adjustments on sonogram to answer clinical questions
- Reviews previous sonogram, and provides oral or written summary of technical findings to physician
- Alerts physician if critical findings are found

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Qualities of a Sonographer

- Intellectual curiosity
- Perseverance
- Ability to conceptualize two-dimensional images into a three-dimensional format
- Quick and analytical mind
- Technical aptitude
- Good physical health

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Qualities of a Sonographer (Cont.)

- Independence and initiative
- Emotional stability
- Communication skills
- Dedication

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Advantages of a Sonography Career

- Specialized education in ultrasound garners respect of other allied health professionals and clinicians
- Provides daily opportunities to use creativity to meet the challenge of performing quality ultrasound examinations on difficult patients

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Advantages of a Sonography Career (Cont.)

- New applications in ultrasound and improvements in instrumentation provide continual challenge
- Flexible schedules and varieties in examinations, equipment, patient personalities
- Certified sonographers have abundant employment opportunities with attractive salaries

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Disadvantages of a Sonography Career

- Some find career overly stressful and demanding
- Changes in medical care and decreased staffing may lead to increased workloads
- Hours of continual scanning may lead to tendinitis, arm and shoulder pain, and back strain
- Working with terminally ill patients can lead to fatigue and depression

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Job Market in Sonography

- Sonography is the top of all allied health imaging salary range
- Demand for certified sonographers exceeds supply nationwide
- Positions at hospital or other health care facilities may include:
  - *Director of Imaging, Technical Director, Supervisor*
  - *Chief Sonographer, Sonographer Educator*
  - *Research Sonographer, Clinical Staff Sonographer, Clinical Instructor*

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Job Market in Sonography (Cont.)

- Sonographers with advanced degrees may serve as Manager, Supervisor, Program Director, Department Head, Dean of Allied Health
- Commercial (Clinical Application Specialist, Director of Education, Marketing Director, Product Design)
- Independent business partners

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Resource Organizations

- American Society of Echocardiography (ASE)
- Society of Diagnostic Medical Sonography (SDMS)
- Society for Vascular Ultrasound (SVU)
- American Institute of Ultrasound in Medicine (AIUM)
- American Registry for Diagnostic Medical Sonography (ARDMS)
- Joint Review Committee on Education in Diagnostic Medical Sonography (JRC-DMS)
- Joint Review Committee on Education in Cardiovascular Technology (JRC-CVT)

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Chapter 2

Essentials of Patient Care for the Sonographer

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Sonographer Obligations

- Sonographers have four main obligations
  - To their patients
  - To their sonologist, department, or institution
  - To the profession
  - To themselves

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Patient-Focused Care

- The most important facet of being a sonographer is the recognition that the patient is the primary focus of our efforts.
- Patient-focused care (PFC) is a national movement designed to recapture the respect and good will of the American public.

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Basic Patient Care

- Vital signs are observable and measurable signs of life.
  - Pulse
  - Respiratory rate
  - Temperature
  - Blood pressure

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Pulse

- Easy and effective way to measure the heart rate.
- Recorded as beats per minute (bpm).
- The beat of the pulse should be evaluated for rate, rhythm, and regularity, as well as for its strength and tension.
- The most common arrhythmias are tachycardia and bradycardia.

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Respiration

- Respiration, or breathing, is the process of inhaling and exhaling air.
- The primary function is to obtain oxygen for use by the body's cells and to eliminate the cells' production of carbon dioxide.
- Normal breathing is quiet, effortless, and regular in rhythm.
- To count respirations, the number of inhalations per minute are noted.

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Respiration (Cont.)

- Dyspnea: Shortness of breath or the feeling of not getting enough air; dyspnea may leave a person gasping.
- Apnea: Spontaneous breathing that stops for any reason. It may be either a temporary starting and stopping at intervals or prolonged.

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Respiration (Cont.)

- Hyperventilation: Rapid breathing in excess of body requirements. It results in an excessive loss of carbon dioxide from the body.
- Respiratory arrest: A life-threatening stoppage of breathing that requires emergency medical assistance.

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Blood Pressure

- The pressure exerted by circulating blood against the walls of the blood vessels.
- The higher number is the systolic pressure, which occurs when the ventricles contract to pump blood to the body.
- The lower number is the diastolic pressure, which occurs near the end of the cardiac cycle when the ventricles are filling with blood.

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Hypertension

- Also known as high blood pressure, hypertension directly increases the risk of coronary heart disease and stroke.
- When blood pressure is high, the arteries may have increased resistance against the flow of blood, causing the heart to pump harder.

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Isolated Systolic Hypertension

- Exists when the systolic pressure is above 140 mm Hg, but the diastolic pressure remains below 90 mm Hg.

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Patient Needs

- Bedpan
- Urinal
- Emesis basin
- Intravenous (IV) infusion tubing
- Nasogastric suction tubing
- Urinary catheter
- Nasal catheter or cannula used for oxygen administration

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Wound Drains and Dressings

- A wound drain is created by inserting a wick inside the wound to provide drainage.
- Do not pull or dislodge the drain when positioning, scanning, or transferring the patient.
- If a wound is actively draining, a gauze pad can be taped over the area until the scan is complete.
- If the wound is within the scanning field, sterile technique is required to scan over the open wound.
- Check with the nursing staff to see if wound precautions are in place.

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Ostomy Patients

- Ostomy patients wear an external bag to collect liquefied fecal matter.
- Changing an ostomy bag requires clean rather than sterile technique.
- The time and date of bag removal and replacement are recorded; the character of the drainage is noted.

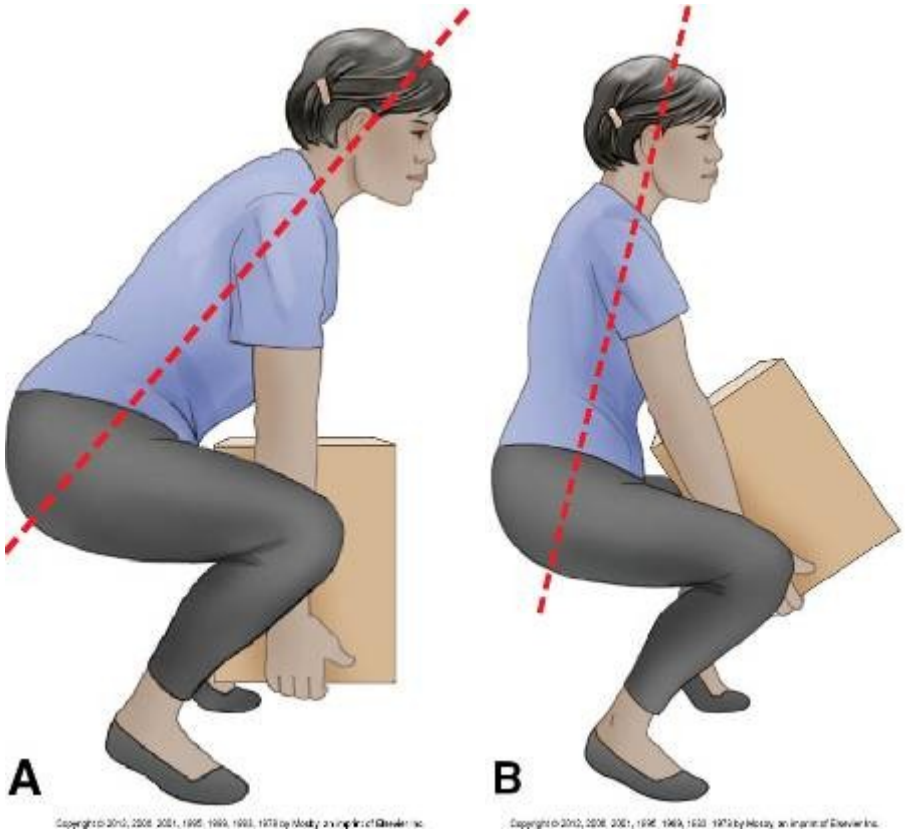
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Patient Transfer Techniques

- Wheelchair transfers
- Stretcher transfers
- Body mechanics
  - Maintain a stable center of gravity
  - Maintain a strong base of support
  - Keep the back straight and objects close to the body
  - Maintain proper body alignment

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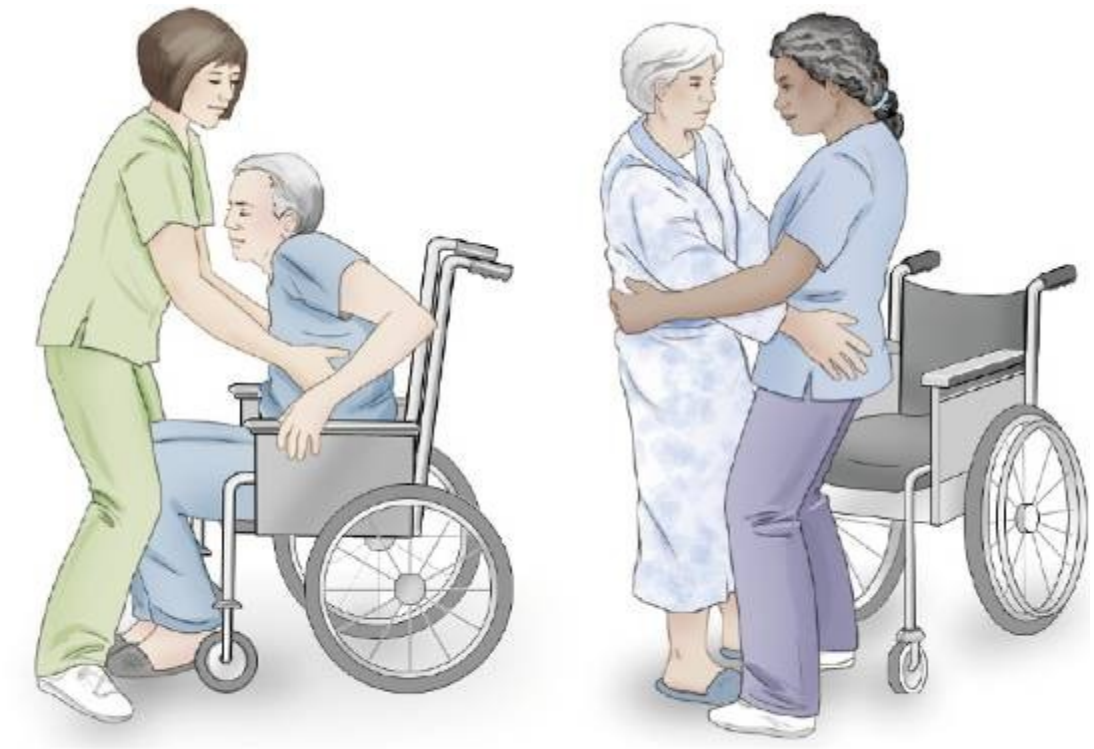
Proper Body Mechanics for Lifting



A, Keep the back straight. Use the large muscles of the thigh, not the small muscles of the back. **B,** Tighten abdominal and pelvic muscles, tuck buttocks in, and keep head and chest up.

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Wheelchair Transfers



A

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B

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Stretcher Transfers


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Using a pulling sheet is recommended whenever transferring patients who cannot help themselves.

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Infection Control

- The Centers for Disease Control and Prevention (CDC) established Standard Precautions and Additional Precautions to expand the Universal Precautions developed in the 1980s.
- The Standard Precautions added additional body fluids and sites to the earlier protocols.

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Infection Control (Cont.)

- Standard Precautions are the basic infection-control guidelines designed to reduce the risks of infection spread through the following three transmission modes:
  - Airborne
  - Droplet
  - Contact

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Infection Control (Cont.)

- Wash your hands whenever you come in contact with the following substances:
  - Blood
  - All body fluids (e.g., secretions, excretions, contaminated items, even if blood is not visible; the only exception is perspiration)
  - Broken skin
  - Mucous membranes (i.e., inside the mouth, nose, and eyelids)
  - Dried blood and/or dried bloody fluids

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Environmental Control

- Keep the work environment as clean as possible.
  - Adequate procedures must be in place for the routine care, cleaning, and disinfection of environmental surfaces, beds, and side rails.
  - All equipment must be cleaned according to institutio n's policies.
  - Equipment includes blood pressure cuffs, basins, bedpans, and wheelchairs.

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Airborne Transmission

- Diseases that spread by airborne transmission include the following:
  - Tuberculosis
  - Measles
  - Chicken pox
  - Shingles

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Droplet Transmission

- Germs that are too heavy to remain airborne can drop quickly.
- Diseases spread via droplet transmission include the following:
  - Mumps
  - Measles (rubella)
  - Whooping cough (pertussis)
  - Pneumonia
  - Meningitis (specific forms)
  - Strep throat

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Contact Transmission

- Germs can spread directly or indirectly by contact with (touching) the germ. The following diseases are spread by contact:
  - Methicillin-resistant *Staphylococcus aureus* (MRSA)
  - *Escherichia coli*
  - Wound infections
  - Influenza
  - Impetigo
  - Pink eye
  - Scabies
  - Hepatitis A

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Methicillin-Resistant *Staphylococcus Aureus*

- MRSA is a form of staphylococcal bacteria.
- Bacteria lives on the skin and nasal passages of one-third of the world's population.
- Resistant to certain broad-spectrum antibiotics

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Methicillin-Resistant *Staphylococcus Aureus* (Cont.)

- Community-acquired MRSA
  - Typically produces skin infections
- Hospital-acquired MRSA
  - Can cause serious lung and bloodstream infections

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Methicillin-Resistant *Staphylococcus Aureus* (Cont.)

- MRSA is primarily spread on the hands of health care workers and infected persons.
- Draining wounds and infected discharge are other routes of transmission.
- Best defense against MRSA in hospital settings is hand washing and the proper use of barrier devices.

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Precautions for Handling Biohazardous Waste

- Refuse that has been contaminated with germs should be bagged and labeled.
  - Includes discarded dressings, used needles, and contents of bedpans or urinals.
- Used sharps must always be placed in puncture-proof containers.

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Bloodborne Transmission

- Bloodborne diseases are those spread when the blood of an infected person comes in contact with the blood of another person.
- Human immunodeficiency virus (HIV), acquired immunodeficiency syndrome (AIDS), and hepatitis are three of the most common diseases spread by bloodborne transmission.

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Nosocomial Infections

- Nosocomial infections are hospital-acquired infections.
- Contracted as a result of medical treatment, nosocomial infections usually manifest within the first 48 hours of treatment.
- Most common forms are urinary tract infections, pneumonia, and surgical incision site infections.

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Nosocomial Infections (Cont.)

- Transmission of nosocomial infections occurs through the following:
  - Direct contact: Person to person
  - Indirect contact: Touching an infected surface or a surface treated with improperly sterilized equipment
  - Droplet infection: Sneezing and coughing
  - Airborne transmission: Sneezing and coughing
  - Common vehicle transmission: Food, water, or medical devices
  - Vector transmission: Contact with or bites from an insect or animal

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Preparing a Sterile Field

- Medical aseptic technique is designed to rid an area or object of pathogenic microorganisms.
- Aseptic technique is commonly used in procedures that involve puncturing the skin or when placing objects into normally sterile body cavities.
- See Figure 3-21 in your textbook.

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Emergency Medical Situations

- Choking
- Cardiopulmonary respiration
- Barrier devices
- Automated external defibrillator (AED)

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Patients with Special Needs

- How can the sonographer best handle these types of patients with special needs?
  - Crying patients
  - Pediatric patients
  - Adolescent patients
  - Elderly patients
  - Culturally diverse patients

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Patient Reactions to Illness

- Denial or disbelief: Patients may avoid, refuse, or forget about needed care or appointments.
- Acceptance: Some patients may become dependent on the health care team as they focus attention on their symptoms and illness.
- Recovery: Depending on how much lifestyle changes as a result of the illness, the patient eventually begins to recover, rehabilitate, or convalesce.

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Patients' Bill of Rights

- Patients have the right to:
  - Be treated with respect
  - Make a treatment choice
  - Refuse treatment
  - Obtain their medical records
  - Privacy of their medical records
  - Informed consent
  - Make decisions about end-of-life care

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Patient Responsibilities

- Patients also have responsibilities
  - Maintain healthy habits
  - Be respectful to providers
  - Be honest with providers
  - Comply with treatment plans
  - Prepare for emergencies
  - Read behind the headlines
  - Make decisions responsibly

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The Health Portability and Accountability Act (HIPPA)

- Three types of standards were created by HIPAA
  - Privacy
  - Security
  - Administrative simplification (e.g., transaction standards)

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HIPAA

- Gives patients more control over their health information
- Sets boundaries on the use and release of health records
- Establishes appropriate safeguards that health care providers and others must achieve to protect the privacy of health information
- Holds violators accountable with civil and criminal penalties that can be imposed if they violate patients' privacy rights
- Strikes a balance between public health and safety and the individual's right to privacy

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HIPAA for Patients

- Enables patients to make informed choices when seeking care and the reimbursement for care, based on how personal health information may be used
- Enables patients to find out how their information may be used and about certain disclosures of their information that have been made
- Generally gives patients the right to examine and obtain a copy of their own health records and to request corrections
- Empowers individuals to control certain uses and disclosures of their health information

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HIPAA for Sonographers

- Putting patient information away after hours
- Taking files out of sight of any lingering staff and custodians
- Setting screensavers on computers for the shortest time possible
- Taking care that any conversations (including phone conversations) are not overheard by other patients
- Removing patient identification from any scans that will be used for publication or presentation

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HIPAA for Sonographers (Cont.)

- Keeping any patient charts filed with the names facing the wall
- Permitting patients to request that students, other observers, medical personnel, and families leave the room during the sonographic examination
- Explaining to patients the hospital or department policies regarding friends and family to view their ultrasound procedure

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